**Code No: 136FT** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year II Semester Examinations, August/September - 2024**
**PRINCIPLES OF COMPUTER COMMUNICATIONS AND NETWORKS**
**(Common to EEE, CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART – A

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | What is computer communication system? | [2] |
| b) | What are the fundamental characteristics for effective data communication? | [3] |
| c) | Define modulation. | [2] |
| d) | How to calculate the speed of the communication channel? | [3] |
| e) | Define line coding schemes. | [2] |
| f) | What are the key differences between analog and digital signals? | [3] |
| g) | Define media access control. | [2] |
| h) | How do guided media differ from unguided media? | [3] |
| i) | What is the role of repeaters? | [2] |
| j) | Write about the Hubs and switches. | [3] |

## PART – B

**(50 Marks)**

2.a) Explain the difference between data communication and computer networking.
b) Differentiate between peer-to-peer and client-server network models. [5+5]

**OR**

3.a) Describe the characteristics of Local Area Network (LAN).
b) What is the purpose of IP addressing in a network? Explain. [5+5]

4. Compare circuit switching, packet switching, and message switching in network communication. [10]

**OR**

5.a) What are the main communication service methods in networks, and how do they differ?
b) List out the advantages of using serial transmission over parallel transmission. [6+4]

6. List out the characteristics of Analog Signals and explain them with necessary diagrammatic representations. [10]

**OR**

7.a) Explain the concept of Frequency Division Multiplexing (FDM) in digital carrier systems.
b) Discuss the impact of channel capacity and bit error rate on digital carrier system performance. [5+5]

8.a) Analyze the advantages of optical fiber over twisted pair and coaxial cable.
b) Examine the access method used by wireless LAN. [5+5]

**OR**

9.a) Explain how the Medium Access Control (MAC) sub-layer manages access to the shared communication medium.
b) How does the data link layer ensure reliable communication between devices? [5+5]

10.a) What are some key differences between SC, LC, and ST connectors in fiber optic networks?
b) List out the common challenges in selecting the right transceiver for specific network needs. [5+5]

**OR**

11.a) Discuss the role of media converters in extending network distances.
b) Explain the difference between wired and wireless network interface cards. [5+5]

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